A symmetric Laurent polynomial $g(z) \in \mathbb{k}[z_1^{\pm 1}, \dots, z_\ell^{\pm 1}]^{S_\ell}$ satisfies the wheel condition if it vanishes whenever any three of the variables z_i, z_j, z_k are in the ratio

$(z_i : z_j : z_k) = (1 : q : qt)$ or $(z_i, z_j, z_k) = (1 : t : qt)$.

If $\ell < 3$, the wheel condition holds vacuously.

e.g. $g(z) = (1 - q\frac{z_1}{z_2})(1 - q\frac{z_3}{z_1})(1 - q\frac{z_1}{z_3})(1 - q\frac{z_2}{z_1})(1 - q\frac{z_3}{z_2})(1 + q\frac{z_3}{z_2})$

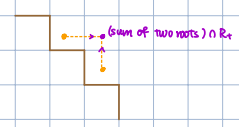
$g(z_1, qz_1, qtz_1) = 0$

Theorem 4.1.1 (Negut): Let $g(z) \in \mathbb{k}[z_1^{\pm 1}, \dots, z_\ell^{\pm 1}]^{S_\ell}$. Then $g(z) \in S_{\hat{\mathfrak{h}}}^{\ell}$ iff $g(z)$ satisfies the wheel condition and vanishes whenever $z_i = z_j$ for $i \neq j$.

* Since g is symmetric, g vanishes whenever $z_i = z_j$ is equivalent to $\prod_{i<j} (z_i - z_j)^2$ dividing $g(z)$.

The wheel condition then applies to $\dfrac{g(z)}{\prod_{i<j}(z_i - z_j)^2}$.

Notation: For $A, B \subseteq R_+$, we set $[A, B] := R_+ \cap (A + B)$.



e.g. $A = \{\alpha_{12}, \alpha_{13}\}$, $B = \{\alpha_{12}, \alpha_{23}\}$

Then $A + B = \{\alpha_{12} + \alpha_{13}, \underbrace{\alpha_{12} + \alpha_{23}}_{\alpha_{13}}, \alpha_{13} + \alpha_{23}, \alpha_{12} + \alpha_{23}\}$ ← only positive root

$\therefore [A, B] = \{\alpha_{13}\}$.

Def 4.1.2: A Catalanimal $H(R_q, R_t, R_{qt}, \lambda)$ is tame if $[R_q, R_t] \subseteq R_{qt}$.

Prop 4.1.3: If $H(R_q, R_t, R_{qt}, \lambda)$ is a tame Catalanimal, then $H \in S_{\hat{\mathfrak{h}}}$. (Hence it corresponds to an element $\mathcal{D}_{\hat{\mathfrak{h}}}(H) \in \mathcal{E}^+$ in the Shuffmann algebra)

Proof: By the equivalent statements in §3.4, $H \in S_{\hat{\mathfrak{h}}}$ iff $g \in S_{\mathfrak{h}}$.

Hence we only need to check if $g(z)$ satisfies the wheel condition and vanishes whenever $z_i = z_j$ for $i \neq j$, and result follows by Theorem 4.1.1.

If we can prove $z^\lambda \prod_{\alpha \in R_+} (1 - z^\alpha) \prod_{\alpha \in R \setminus R_q} (1 - q z^\alpha) \prod_{\alpha \in R \setminus R_t} (1 - t z^\alpha) \prod_{\alpha \in R_{qt}} (1 - qt z^\alpha)$ vanishes under those conditions, then $g(z) = \sum_{w \in S_\ell} w\Big(z^\lambda \prod_{\alpha \in R_+} (1 - z^\alpha) \prod_{\alpha \in R \setminus R_q} (1 - q z^\alpha) \prod_{\alpha \in R \setminus R_t} (1 - t z^\alpha) \prod_{\alpha \in R_{qt}} (1 - qt z^\alpha)\Big)$ also vanishes under those conditions.

(i) If $z_i = z_j$ for $i \neq j$, WLOG, say $i < j$, then $\alpha_{ij} \in R_+$ and hence $1 - q z^{\alpha_{ij}} = 0$. Hence $\prod_{\alpha \in R_+} (1 - z^\alpha) = 0$ whenever $z_i = z_j$ for some $i \neq j$.

(ii) Suppose $(z_i : z_j : z_k) = (1 : q : qt)$.

If $\alpha_{ij} \notin R_q$, then $1 - q z^{\alpha_{ij}} = 1 - q \overbrace{\frac{z_i}{z_j}}^{\frac{q z_i}{q z_j} = 1} = 0$ and hence $\prod_{\alpha \in R \setminus R_q} (1 - q z^\alpha) = 0$.

If $\alpha_{jk} \notin R_t$, then $1 - t z^{\alpha_{jk}} = 1 - t \underbrace{\frac{z_j}{z_k}}_{t \cdot \frac{q z_i}{q t z_i} = 1} = 0$ and hence $\prod_{\alpha \in R \setminus R_t} (1 - t z^\alpha) = 0$.

If $\alpha_{ij} \in R_q$ and $\alpha_{jk} \in R_t$, then $\alpha_{ik} = \alpha_{ij} + \alpha_{jk} \in [R_q, R_t] \subseteq R_{qt}$, and $1 - qt \frac{z_i}{z_k} = 1 - qt \frac{z_i}{(qtz_i)} = 0$ and hence $\prod_{\alpha \in R_{qt}} (1 - qt z^\alpha) = 0$

(iii) Suppose $(z_i : z_j : z_k) = (1 : t : qt)$.

If $\alpha_{ij} \notin R_t$, then $1 - t z^{\alpha_{ij}} = 1 - t \overbrace{\frac{z_i}{z_j}}^{\frac{t z_i}{t z_j} = 1} = 0$ and hence $\prod_{\alpha \in R \setminus R_t} (1 - t z^\alpha) = 0$.

If $\alpha_{jk} \notin R_q$, then $1 - q z^{\alpha_{jk}} = 1 - q \underbrace{\frac{z_j}{z_k}}_{q \cdot \frac{t z_i}{q t z_i} = 1} = 0$ and hence $\prod_{\alpha \in R \setminus R_q} (1 - q z^\alpha) = 0$.

If $\alpha_{ij} \in R_t$ and $\alpha_{jk} \in R_q$, then $\alpha_{ik} = \alpha_{ij} + \alpha_{jk} \in [R_q, R_t] \subseteq R_{qt}$, and $1 - qt \frac{z_i}{z_k} = 1 - qt \frac{z_i}{(qtz_i)} = 0$ and hence $\prod_{\alpha \in R_{qt}} (1 - qt z^\alpha) = 0$

$\therefore g(z) \in S_{\mathfrak{h}}$ and result follows. $\square$